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# CLIMATE & WEATHER SYSTEMS

## Group - 15

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# Probabilistic Monsoon Rainfall Prediction for Agricultural Planning

## Real-world context of the problem :-

- Gujarat Farmers decide crop sowing windows based on mansoon outloolk
- Drought years (<750mm ) cause direct agricultural and economic loss
- 1901-2015 data shows ~31% of years fall below the drought thresold

## What M-3 does :-

- Translates the M2 mathematical framework into a running simulation
- Uses the simplest possible non-random forecast as a reference point
- Establishes the 68.7% accuracy floor that M4 must beat

# Project System Overview

# End-to-End Code Pipeline :-

**STEP 1: Load & Filter Data**

df_raw [SUBDIVISION == "GUJARAT REGION"]

→ 115 years, Jun–Sep monsoon column

**STEP 2: Log Transformation (M2 § 2.2)**

Y = log (monsoon)

Reduces skewness → justifies Gaussian

**STEP 3: Fit Parameters**

μ_Y = mean (Y) = 6.725

σ_Y = std (Y) = 0.34

**STEP 4: Deterministic Baseline**

X = μ_Y for ALL years (constant)

→ ρ = Corr (X, Y) ≈ 0

**STEP 5: Conditional Distribution ( M2 § 4)**

μ_{Y | x} = μ_Y + ρ·(σ_Y/σ_X)·(x−μ_X)

= 6.725 (no shift, ρ = 0)

σ_{Y | x} = σ_Y·√(1−ρ²) = 0.34

**STEP 6: Exceedance Probability (M2 §5)**

P (Y > t | X = x) = 1 − Φ ((t − μ_{Y | x}) / σ)

= 0.622 (same every year)

**STEP 7: Decision Rule**

P ≥ 0.50 → SOW

P < 0.50 → HOLD

Result :- Always Sow (P = 0.622 > 0.50)

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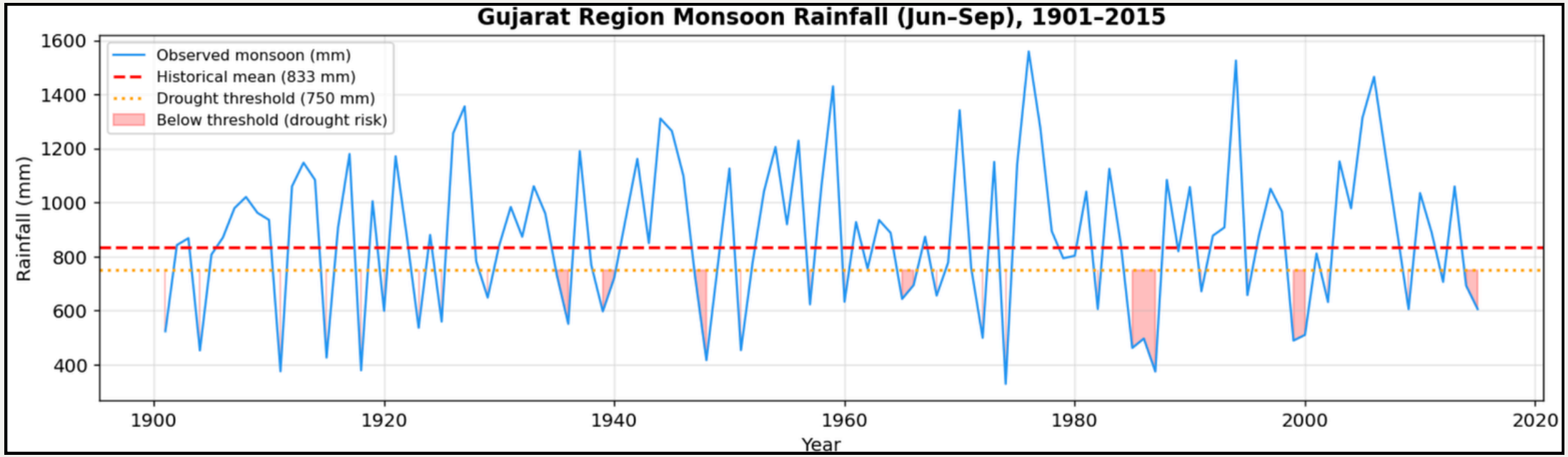
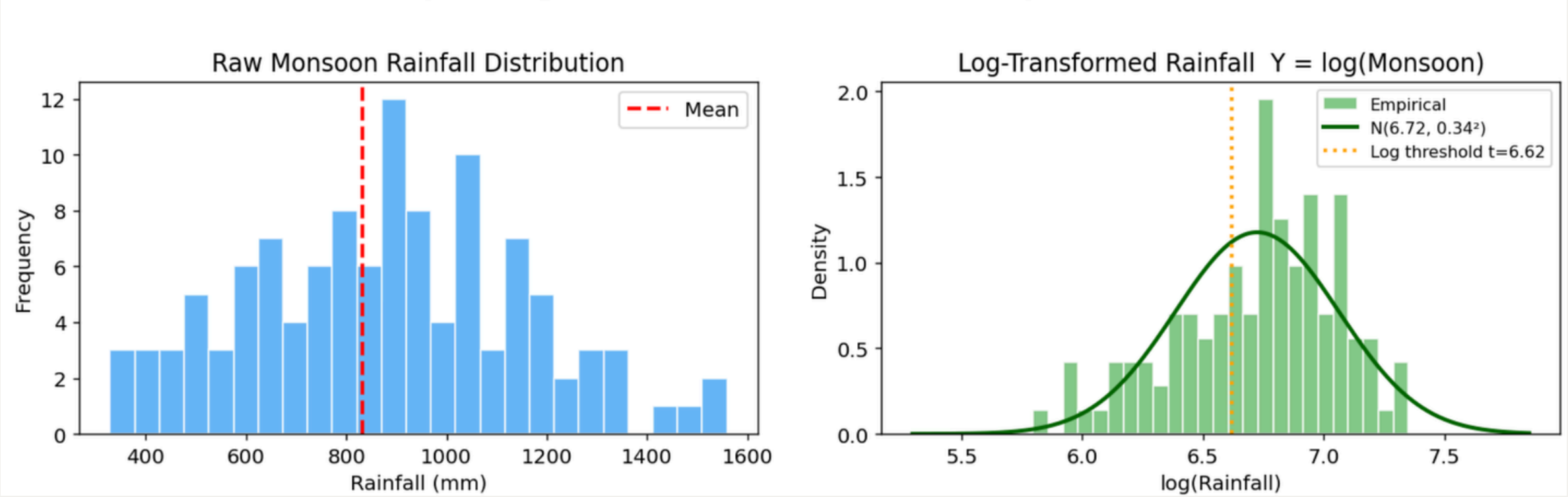
# Gujarat Monsoon Rainfall (1901–2015)




| Curve Color | Forecast Value | Meaning |
|---|---|---|
| Blue | (x = μ_X - σ) | Below-average forecast → probability distribution shifts left |
| Green | (x = μ_X) | Mean forecast → distribution remains centered |
| Orange | (x = μ_X + σ) | Above-average forecast → distribution shifts right |
| Red | (x = μ_X + 1.5σ) | Strong forecast signal → large rightward shift in distribution |

# Marginal Distribution & Log Transformation

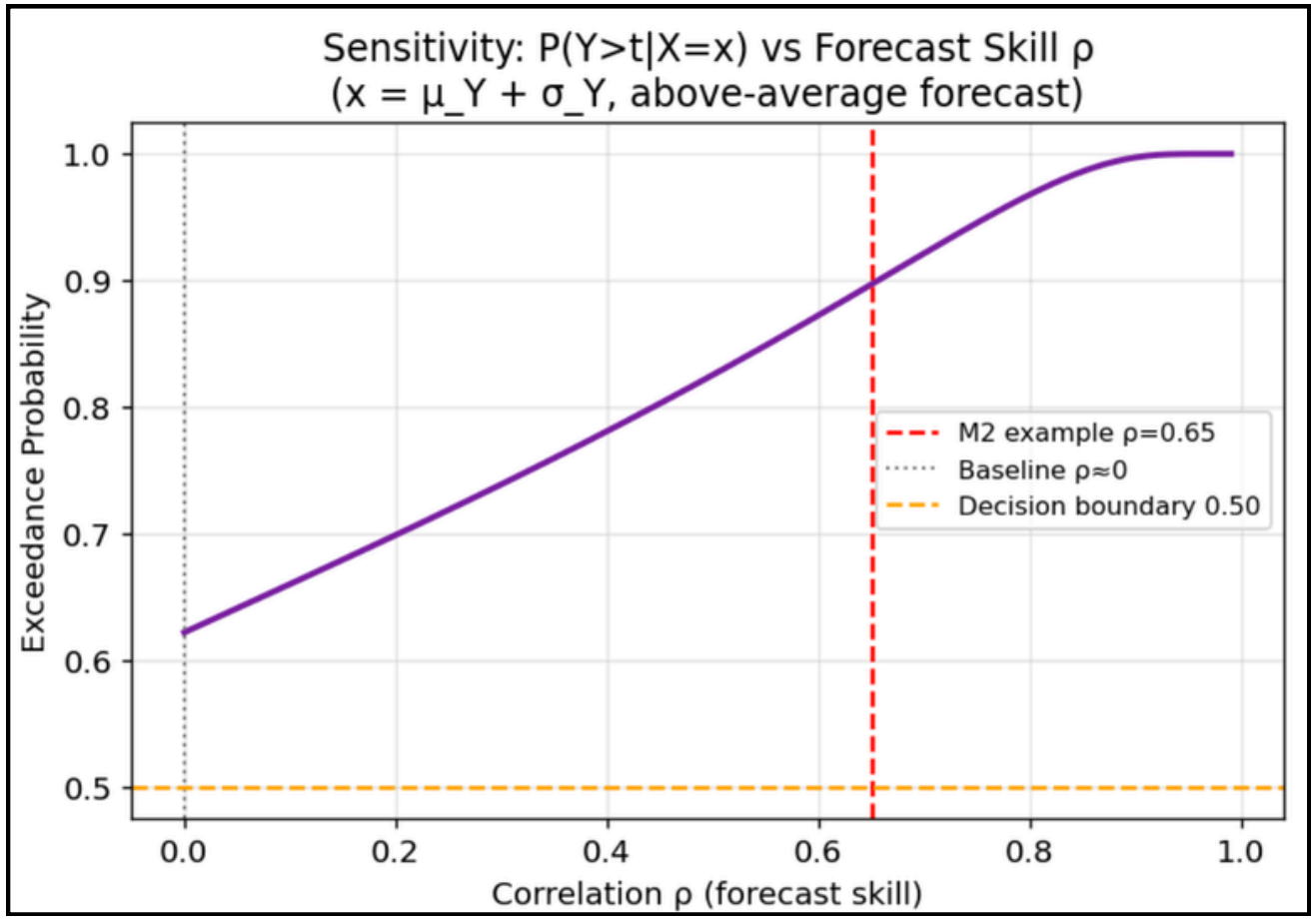



- Raw rainfall is right-skewed - long tail toward high values
- Cannot directly apply Gaussian models to skewed data
- Red dashed line = historical mean ~ 833 mm

**$Y = \log(\text{Monsoon}) \rightarrow Y \sim N(\mu_Y = 6.725, \sigma_Y^2 = 0.34^2)$**

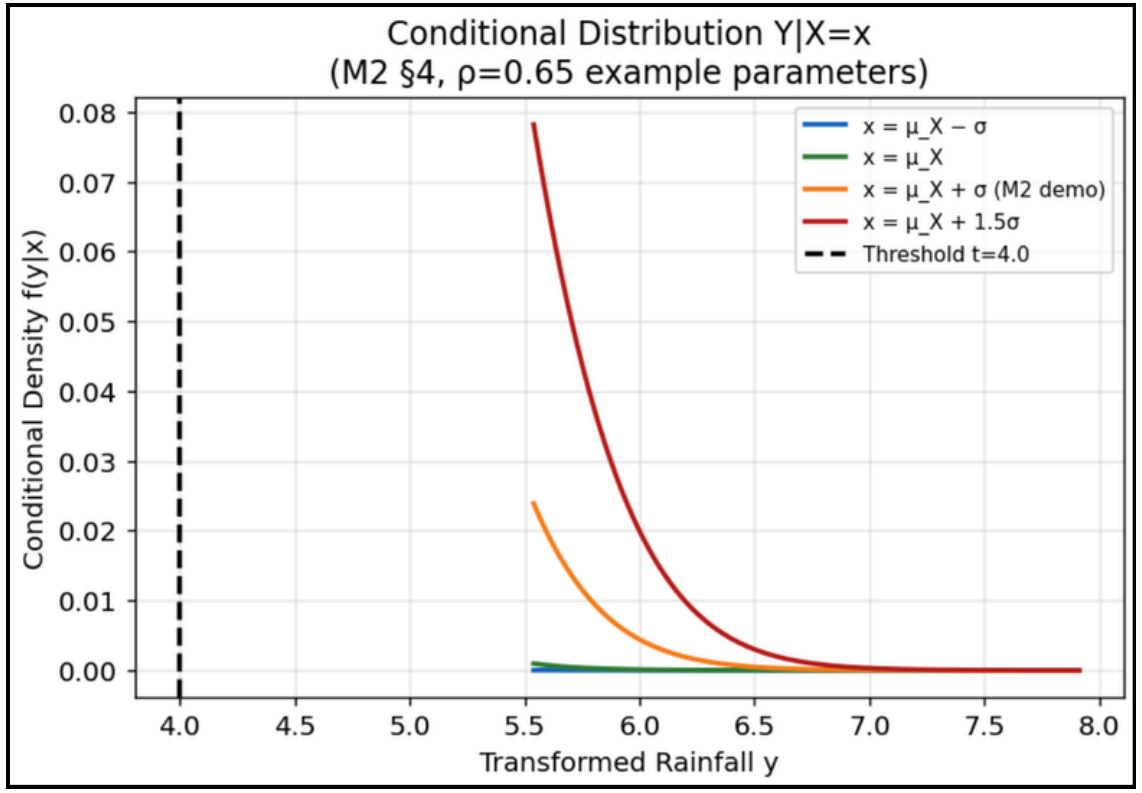
- After Y = log(Monsoon), distribution becomes approximately bell-shaped
- Fitted curve: $N(6.72, 0.34^2)$ - reasonable Gaussian fit
- Orange dotted line = log threshold t = 6.62
- This justifies the bivariate normal assumption in Milestone 2 and 3

# How Forecast Skill ρ Changes Everything




- At ρ=0 the forecast adds zero information. The curve never crosses the decision boundary for the baseline — we always SOW regardless of any signal.

| Point on Graph | Explanation |
|---|---|
| Grey dotted line at ρ = 0 | Baseline case — exceedance probability P = 0.622 remains constant |
| Rising curve | As ρ increases, forecast skill improves and probability moves closer to 1 or 0 |
| Red dashed line at ρ = 0.65 | Example value from Milestone 2 showing moderate forecast skill |
| Orange horizontal line at P = 0.50 | Decision boundary separating SOW vs HOLD decisions |




- With ρ=0.65, each different forecast value shifts the conditional mean. With ρ=0 (baseline), all four curves would be identical — no discrimination.

| Curve Color | Forecast Value | Meaning |
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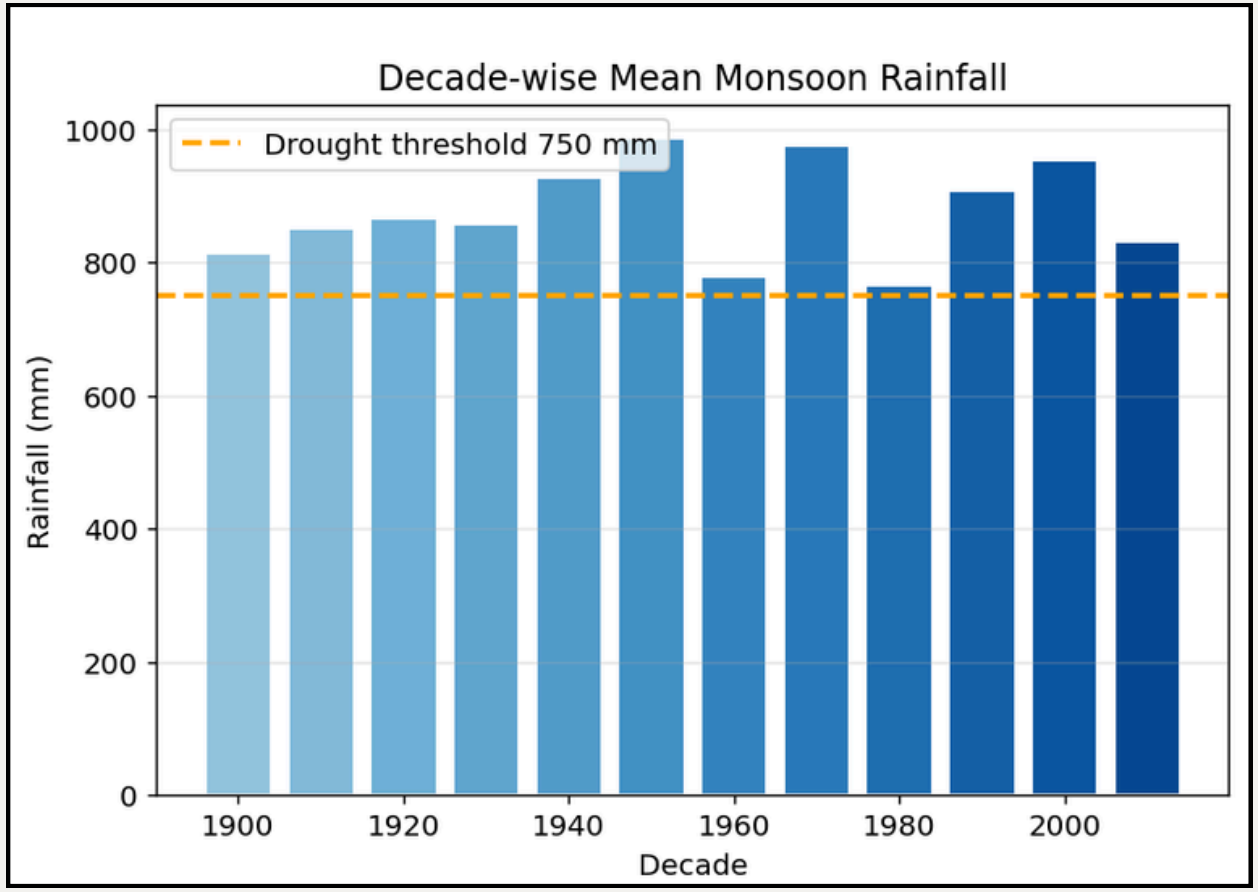
# Replicating the M-2 and 3 Numerical Illustration




Given :- $\mu_X=3.2$, $\sigma_X=0.8$, $\mu_Y=3.0$,

$\sigma_Y=1.0$, $\rho = 0.65$

$\mu_\{Y \mid x\} = 3.0 + 0.65 \times (1.0 / 0.8) \times (4.0 - 3.2)$

$= 3.0 + 0.65 = 3.65$

$\sigma^2_\{Y|x\} = 1.0^2 \times (1 - 0.65^2) = 0.5775$

$\sigma_\{Y|x\} = 0.76$

$z = (4.0 - 3.65) / 0.76 = 0.46$

$P(Y > 4 \mid X = 4) = 1 - \Phi(0.46)$

$= 1 - 0.677$

$= 0.323$

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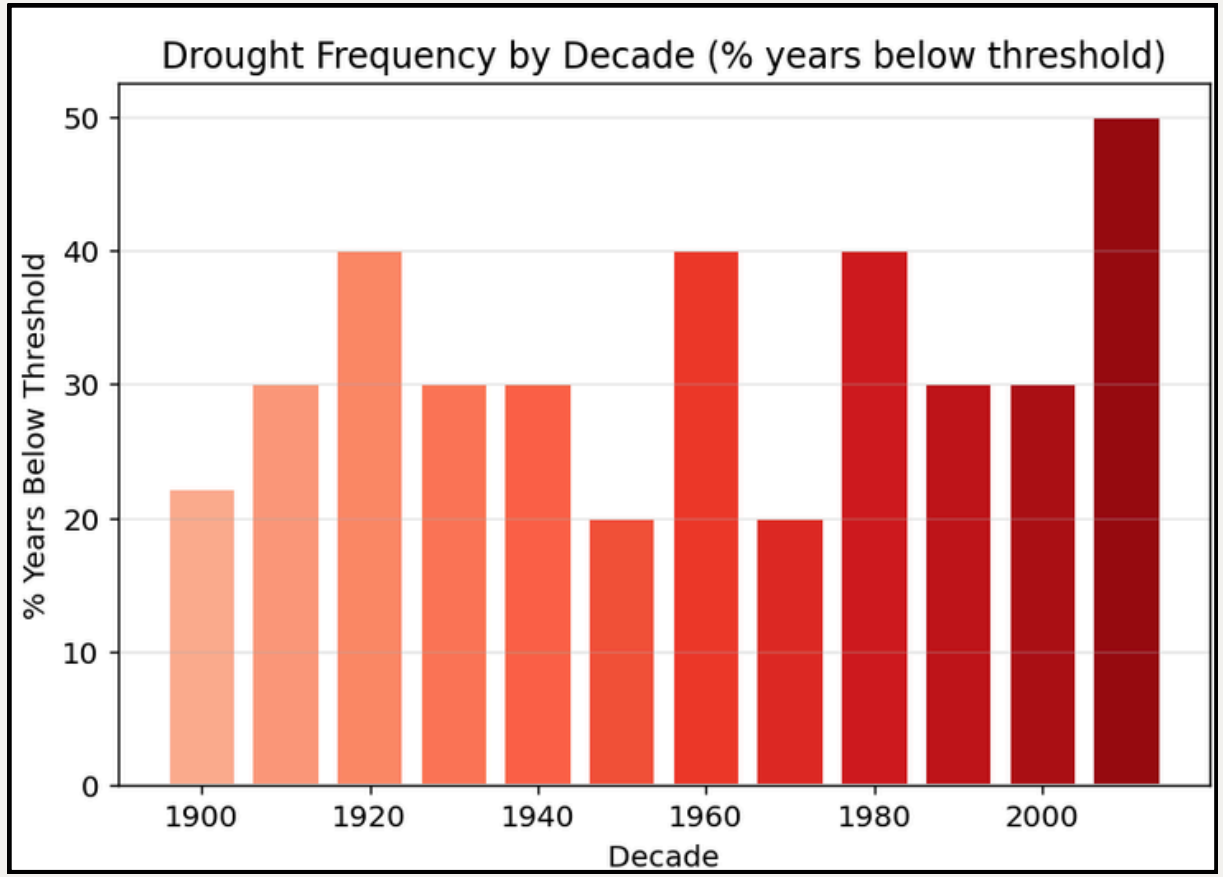



- Decision boundary (P = 0.50) guides SOW vs HOLD.

- Decadal drought frequency highlights rainfall variability.
- Probability curve links forecast strength to rainfall risk.

| **Feature** | **Explanation** |
|---|---|
| S-shaped curve | Classic CDF-complement shape — higher x leads to higher probability P(Y > t) |
| Green dashed line at x = 4 | Example point from Model M2 → P = 0.323 |
| Grey dotted line at x = 3.2 | Baseline forecast ($\mu_x$) → P ≈ 0.16 |
| Orange horizontal line at P = 0.50 | Decision boundary — probability curve crosses around x ≈ 4.3 |

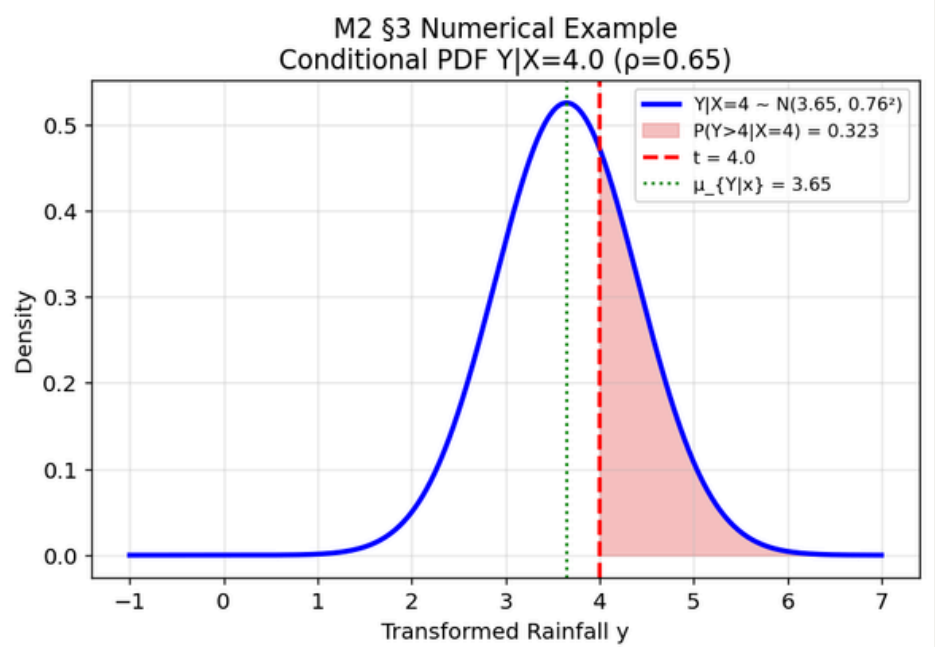
# Replicating the M2 Numerical Illustration

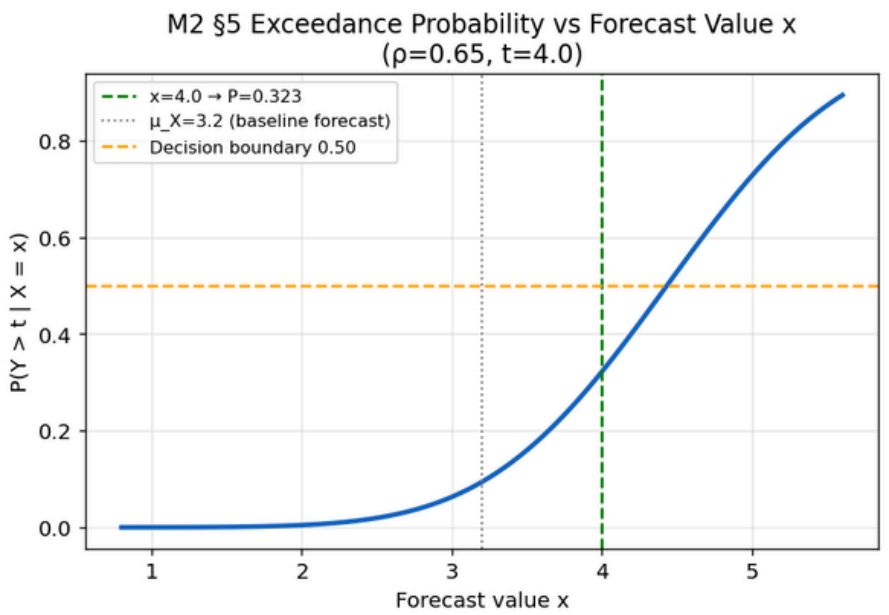
```
=========================================================
  μ_{Y|x} = 3.0 + 0.65×(1.0/0.8)×(4.0−3.2) = 3.6500
  σ²_{Y|x} = 1.0²×(1−0.65²)              = 0.5775
  σ_{Y|x}  =                                 0.7599
  z        = (4.0−3.65)/0.76               = 0.4606
  Φ(z)     =                                  0.6774
  P(Y>4|X=4)=                                 0.3226  (M2 ≈ 0.323 ✓)
```

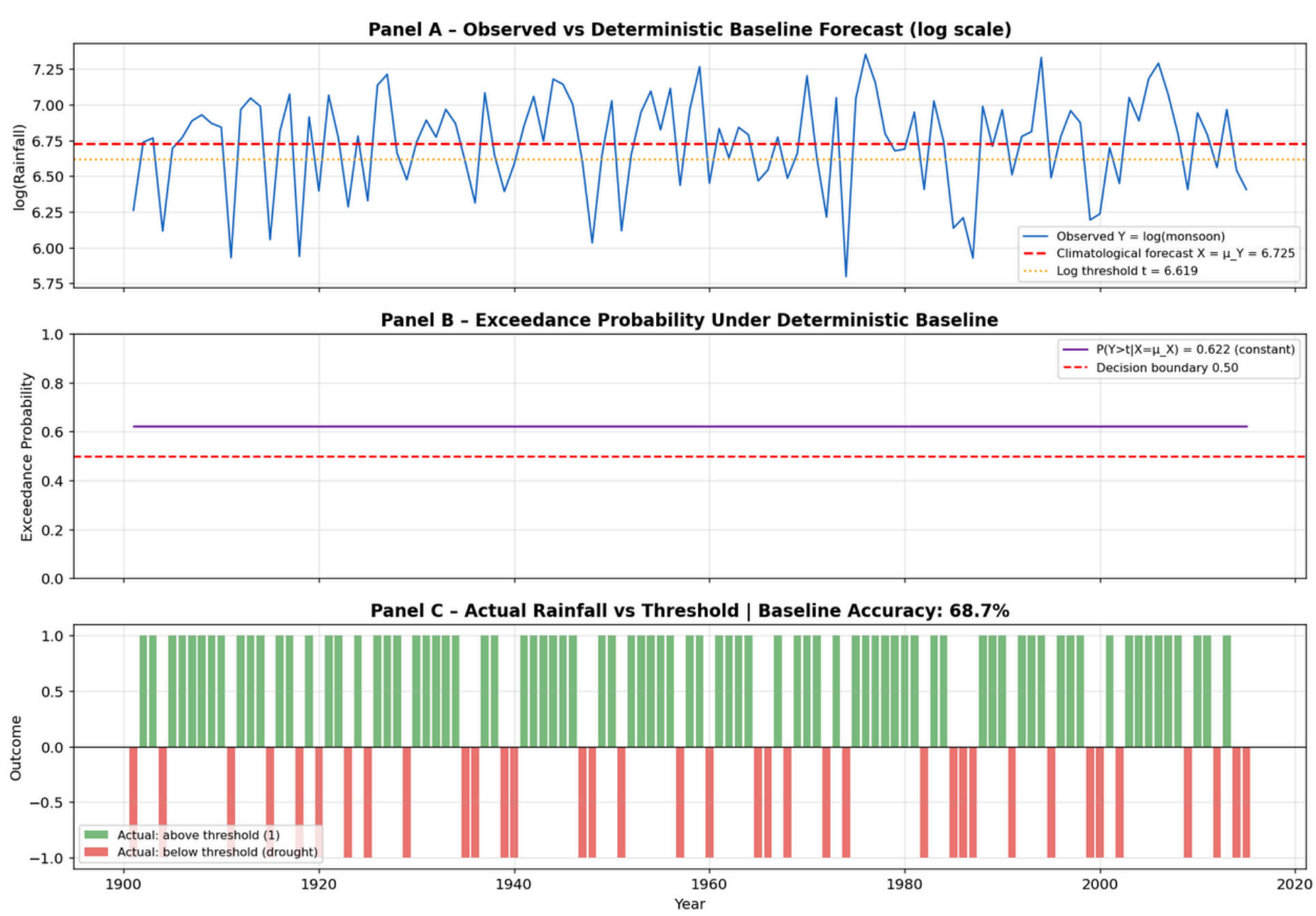
```python
# Left: Conditional PDF for M2 example
yv2 = np.linspace(mu_Y_ex - 4*sig_Y_ex, mu_Y_ex + 4*sig_Y_ex, 500)
pdf_cond = norm.pdf(yv2, mu_cond_ex, sig_cond_ex)
axes[0].plot(yv2, pdf_cond, "b-", lw=2.5, label=f"Y|X=4 ~ N({mu_cond_ex:.2f}, {sig_cond_ex:.2f}²)")
axes[0].fill_between(yv2, pdf_cond, where=(yv2 > t_ex), color="#EF9A9A", alpha=0.6,
                     label=f"P(Y>4|X=4) = {P_ex:.3f}")
axes[0].axvline(t_ex, color="red", lw=2, ls="--", label="t = 4.0")
axes[0].axvline(mu_cond_ex, color="green", lw=1.5, ls=":", label=f"μ_{{Y|x}} = {mu_cond_ex:.2f}")
axes[0].set_title("M2 §3 Numerical Example\nConditional PDF Y|X=4.0 (ρ=0.65)")
axes[0].set_xlabel("Transformed Rainfall y"); axes[0].set_ylabel("Density")
axes[0].legend(fontsize=9); axes[0].grid(alpha=0.3)

# Right: P(Y>t|X=x) as x varies
x_range = np.linspace(mu_X_ex - 3*sig_X_ex, mu_X_ex + 3*sig_X_ex, 300)
P_x_range = []
for xr in x_range:
    mc = conditional_mean(xr, mu_Y_ex, mu_X_ex, sig_Y_ex, sig_X_ex, rho_ex)
    sc = conditional_std(sig_Y_ex, rho_ex)
    P_x_range.append(exceedance_prob(t_ex, mc, sc))
P_x_range = np.array(P_x_range)

axes[1].plot(x_range, P_x_range, color="#1565C0", lw=2.5)
axes[1].axvline(x_ex, color="green", lw=1.5, ls="--", label=f"x={x_ex} → P={P_ex:.3f}")
axes[1].axvline(mu_X_ex, color="gray", lw=1.2, ls=":", label=f"μ_X={mu_X_ex} (baseline forecast)")
axes[1].axhline(DECISION_THRESHOLD, color="orange", lw=1.5, ls="--", label="Decision boundary 0.50")
axes[1].set_title("M2 §5 Exceedance Probability vs Forecast Value x\n(ρ=0.65, t=4.0)")
axes[1].set_xlabel("Forecast value x"); axes[1].set_ylabel("P(Y > t | X = x)")
axes[1].legend(fontsize=9); axes[1].grid(alpha=0.3)
```






The 68.7% accuracy is not skill — it is the base rate. A model that says "SOW every year" will naturally be right 68.7% of the time because 68.7% of years are non-drought years.

# Decadal Analysis

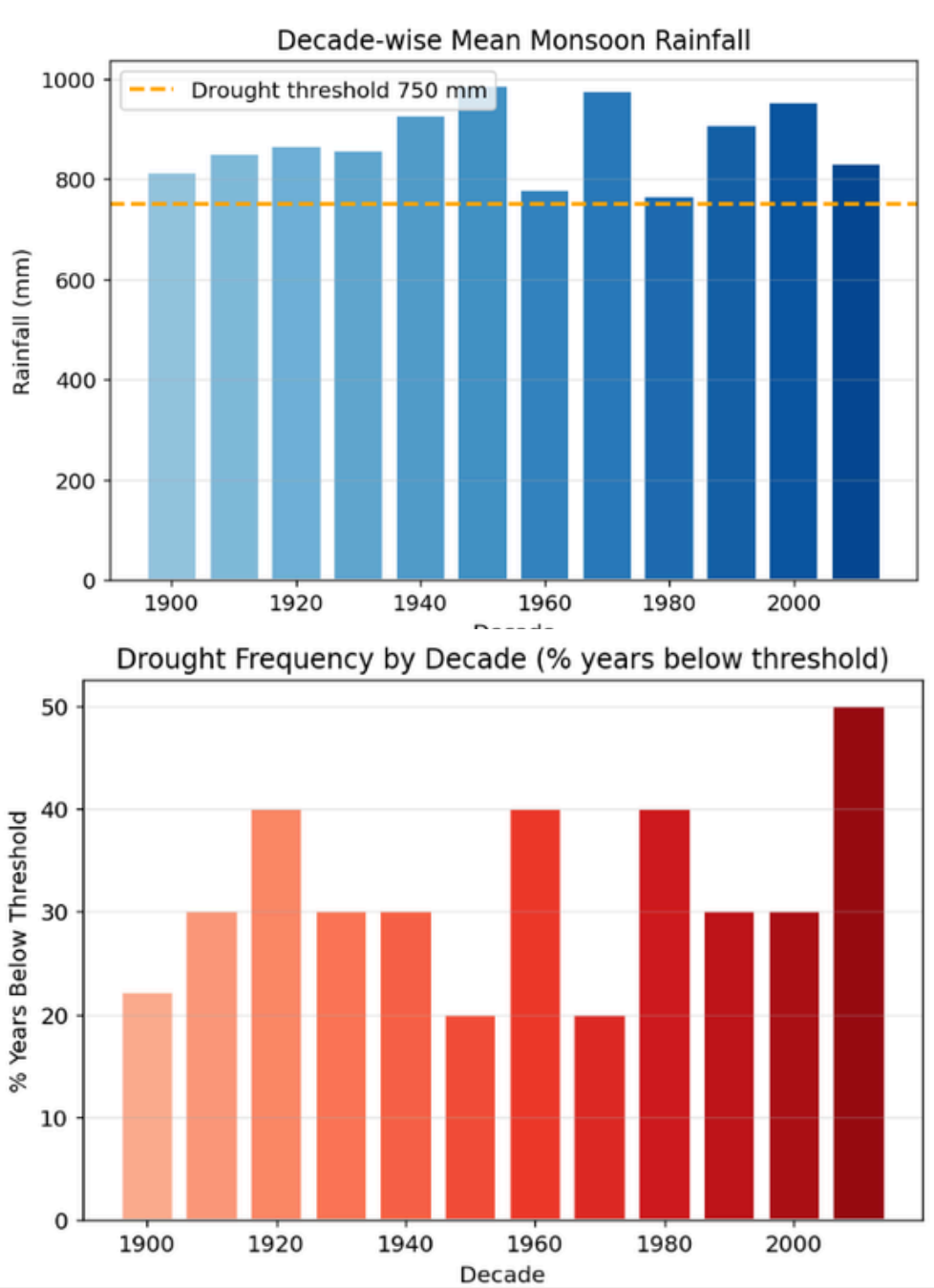
The baseline uses a single fixed μ_Y across all 115 years. But drought frequency has risen from 22% in 1900s to 50% in 2010s. A non-stationary model that adjusts μ_Y over time would be far more appropriate — this is a planned refinement for M4.

```
df_india["DECADE"] = (df_india["YEAR"] // 10) * 10
decade_stats = df_india.groupby("DECADE").agg(
    mean_rain=("MONSOON", "mean"),
    drought_yrs=("actual_above_t", lambda x: (x==0).sum()),
    total_yrs=("actual_above_t", "count")
).reset_index()
decade_stats["drought_pct"] = decade_stats["drought_yrs"] / decade_stats["total_yrs"] * 100

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
bars = axes[0].bar(decade_stats["DECADE"], decade_stats["mean_rain"], width=8,
                   color=plt.cm.Blues(np.linspace(0.4, 0.9, len(decade_stats))), edgecolor="white")
axes[0].axhline(threshold_raw, color="orange", lw=2, ls="--", label=f"Drought threshold {threshold_raw:.0f} mm")
axes[0].set_title("Decade-wise Mean Monsoon Rainfall"); axes[0].set_xlabel("Decade"); axes[0].set_ylabel("Rainfall (mm)")
axes[0].legend(); axes[0].grid(axis="y", alpha=0.3)

axes[1].bar(decade_stats["DECADE"], decade_stats["drought_pct"], width=8,
            color=plt.cm.Reds(np.linspace(0.3, 0.9, len(decade_stats))), edgecolor="white")
axes[1].set_title("Drought Frequency by Decade (% years below threshold)")
axes[1].set_xlabel("Decade"); axes[1].set_ylabel("% Years Below Threshold")
axes[1].grid(axis="y", alpha=0.3)

plt.suptitle("Step 5 – Decadal Analysis: Baseline Model on Historical Data", fontweight="bold", y=1.02)
plt.tight_layout(); plt.savefig("fig6_decadal.png", bbox_inches="tight"); plt.show()
```

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# What the Baseline Cannot Do — And How M4 Will Improve It

## Limitations of M3 Baseline:

| Limitation | Effect |
|---|---|
| Always SOW | Zero drought warnings issued in 115 years |
| Stationary μ_Y | Misses the post-2000 drying trend |
| Gaussian tails | Underestimates extreme drought risk |
| Accuracy = base rate | 68.7% is not a skill score |

## M4 Plan:

- Introduce a randomized/probabilistic forecast with $\rho > 0$
- $P(Y > t \mid X = x)$ will now vary year by year
- Model will sometimes say HOLD — catching drought years
- Evaluate using: CRPS, ROC-AUC, Brier Score — not just accuracy
- Test non-stationary μ_Y using rolling window estimation

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# Summary

PIPELINE COMPLETE

Gujarat data →

log transform →

params fitted →

exceedance →

decision

MATHEMATICAL MODEL VERIFIED

M2 example

$P(Y>4|X=4)=0.323$

confirmed in code output

BASELINE ESTABLISHED

Accuracy: 68.7%

$P(Y>t) = 0.622$

$\rho \approx 0$

Always SOW

**Thank you!**

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